

Hybrid quantum-classical campaign selection

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01 Predict → 02 Select → 03 Simulate

100 probabilities

Boosting + quantum SVM
300 labelled customers

Top 10 candidates

Six customer IDs

QAOA optimization
Exact classical comparison

Within resource limits

Campaign probability

Six independent Ry rotations
Exact + sampled validation

P(at least 3 subscribe)

Stage 1: Hybrid soft voting improves validation probabilities

A hybrid 50:50 soft vote combines classical shallow boosting with the calibrated quantum SVM

The resulting score outperforms both classical baseline and quantum-only SVM

Model	ROC-AUC ↑	AP ↑	Brier ↓
Logistic regression	0.555	0.174	0.110
Shallow boosting	0.610	0.235	0.103
Quantum SVM	0.632	0.199	0.101
50:50 hybrid	**0.645**	**0.255**	**0.099**

Quantum branch

Omit "campaign“ variable
Four PCA components
Input scale 0.25

ZZ feature map, 2 repetitions
QSVC + sigmoid calibration

Refit on all 300 training customers, then score and rank the 100 test customers.

3-fold validation repeated for five seeds

Stage 2: Six customers maximize expected net value

Ten candidates define 210 possible six-customer subsets

Business objective

Maximize $\Sigma\, x_i\,(p_i\, v_i - c_i)$

p_i: subscription probability v_i: deposit value
c_i: contact cost x_i: select or omit

Constraints

Exactly 6 customers
Contact cost ≤ 25
Capacity units ≤ 22

Quantum objective on 10 qubits

Minimize $-\Sigma\, u_i\, x_i + 1.1\,(\Sigma\, x_i - 6)^2$

Stage 2: Every depth samples the optimal selection

Net value 24.30 with zero gap. Selected cost 12/25 and capacity 14/22

Transpiled. 10 logical qubits. Three starts, max 1,000 evaluations/start. 1,024 shots per p.

QAOA	P(feasible)	P(optimum)	Depth*	CX gates*	Time, s
p=1	53.74%	0.318%	**59**	**90**	**1.83**
p=2	60.83%	0.455%	94	180	14.60
p=3	**61.86%**	**0.850%**	129	270	29.48

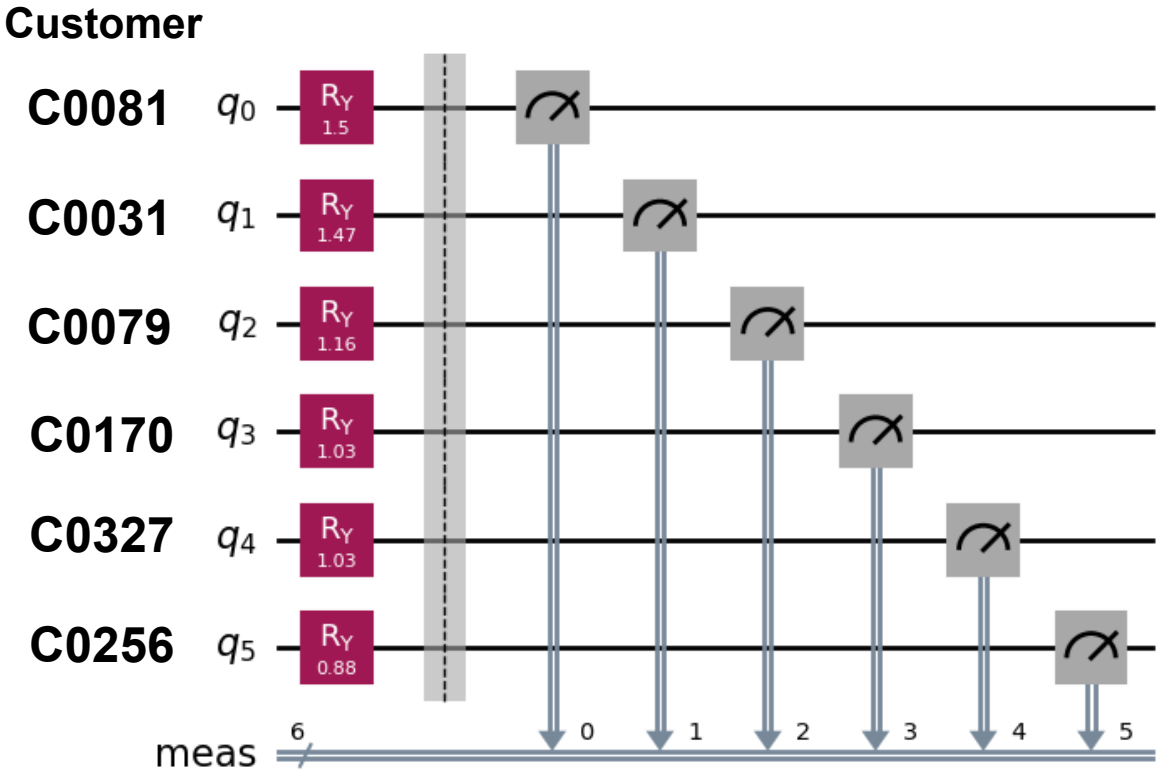
p=3 improves sampling probabilities at 3× the CX count of p=1.

Exact enumeration checks all 210 subsets. All three QAOA depths sample the optimum.

Submitted p=3, original/transpiled: depth 44/129, 1Q gates 70/315, 2Q gates 135/270, SWAPs 0.

Stage 3: About a 28% chance of three subscriptions

Each selected customer contributes one qubit with P(1) equal to their Stage 1 probability



$\theta_i = 2 * \arcsin(\sqrt{p_i})$

Success: at least three measured

Method	P(≥3)	Error, pp
Exact reference	27.59%	—
Quantum sampling	**27.73%**	**0.14**
Classical MC	30.47%	2.88

Each uses 1,024 draws. SE ≈ 1.40 pp.

1.88 expected subscribers

The three-subscription target remains unlikely under the model.